Characterization of Nanoparticle Interactions in Peripheral Neural Lesion Models

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Introduction

- Neural lesions induce transdifferentiation of Schwann cells (SCs) to a repair phenotype, wherein repair SCs recruit and participate with macrophages in myelin clearance
- Inadequate clearance results in failed nerve repair

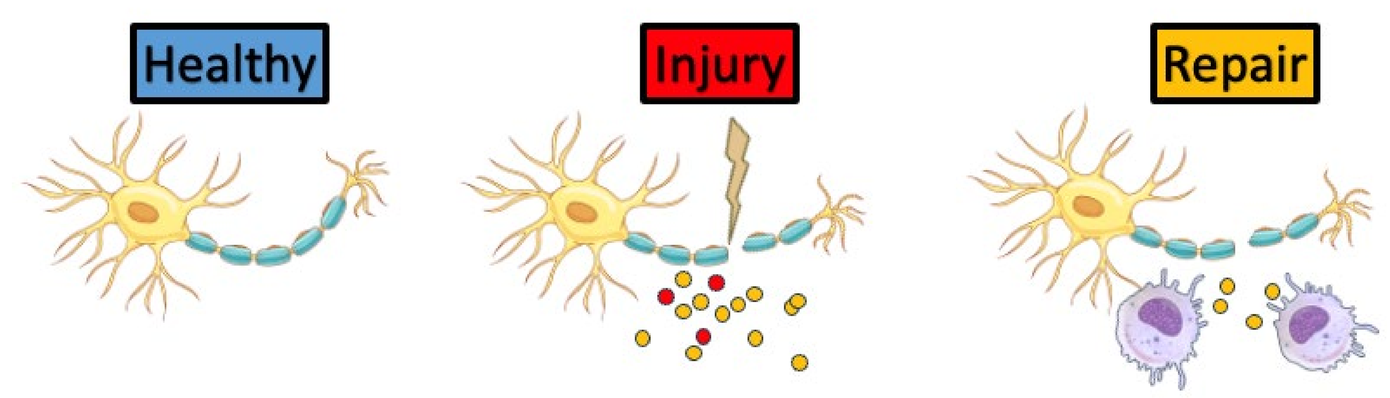


Figure 1: Neural injury model. Disrupted myelin sheath becomes myelin debris and promoting SC autophagy. SCs secrete cytokines that recruit macrophages

- Nanoparticles (NPs) can promote repair, enhanced drug delivery and tissue specificity, and tunable release kinetics
- NPs are limited by trafficking issues, resulting in poor target accumulation
- Chondroitin sulfate (CS) NPs may be immunomodulatory, but their effects on SCs are unknown
- NP Interactions in neural lesion contexts are not well understood

OBJECTIVE

Provide insight into how NP therapies could be leveraged to provide neural repair by defining dose-dependent uptake dynamics, clarifying competitive interactions during myelin clearance

Methods

MTS METABOLIC ACTIVITY ASSAY

Treatment (μg/mL):
Blank NPs: 1200, 600, 300, 150
CS-NPs: 1200, 600, 300, 150
DCS-NPs: 1200, 600, 300, 150

Incubation: 24 hours: BNPs, CS-NPs, DCS-NPs.
72 hours: CS-NPs, DCS-NPs

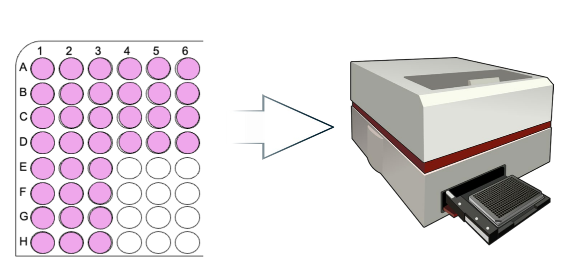


Figure 2: MTS Assay, Cytotoxicity

NP UPTAKE ASSAY

BNP Treatments (μg/mL):
1200, 600, 300, 150

Fluorescent Staining: DAPI (461 nm), CellMask Green (488 nm), Alexa Fluor (647 nm)

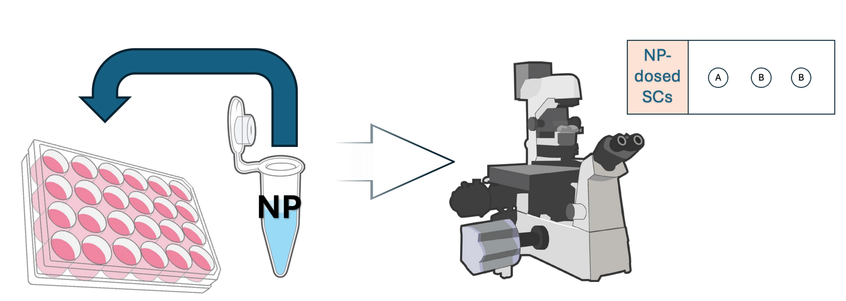


Figure 3: Fluorescent microscopy, BNP uptake

COMPETITIVE UPTAKE

Treatments:
CS-NPs & DCS-NPs:600 μg/mL
Myelin Debris (MD): 300 μg/mL, 600 μg/mL

Incubation: 0-12 hrs, imaging every hour

Fluorescent Staining:
Cell Membrane: CellBrite Green (488 nm),
NPs: Alexa Fluor (647 nm),
Myelin: Fluoromyelin (555 nm)

Results

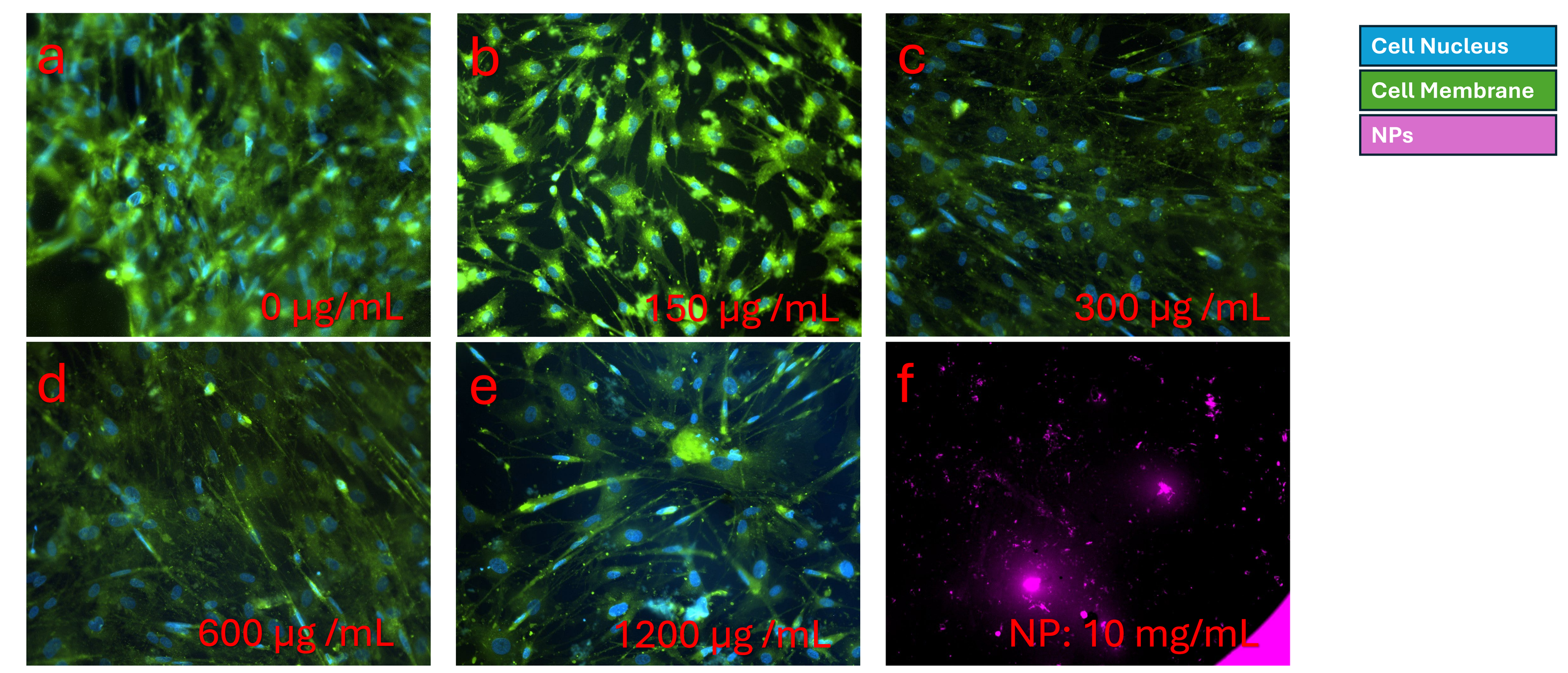
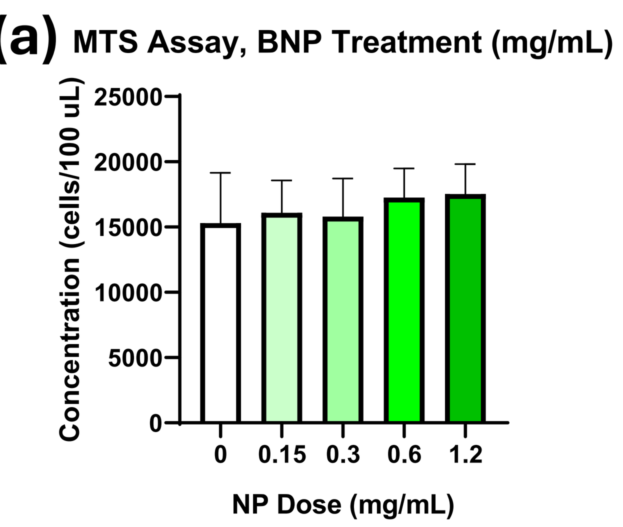
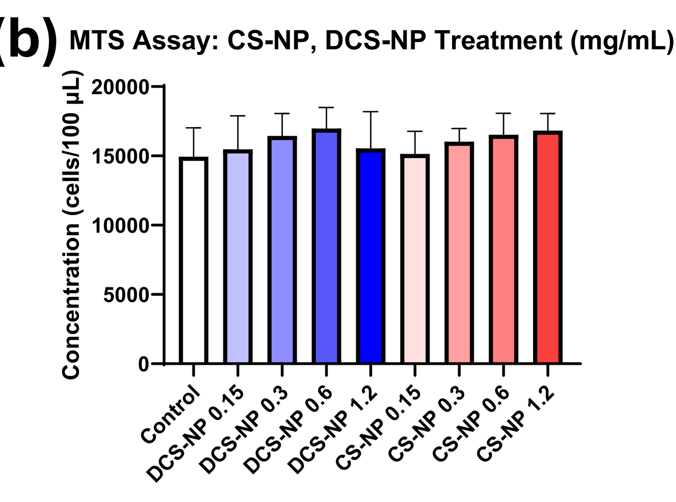


Figure 4: hSC BNP Uptake Experiment. Panels (a-e) show hSCs dosed at 0 μg/mL (a), 150 μg/mL (b), 300 μg/mL (c), 600 μg/mL (d), and 1200 μg/mL (e). Panel f shows a 10 mg/mL BNP only control. Images taken at 20x magnification on inverted microscope

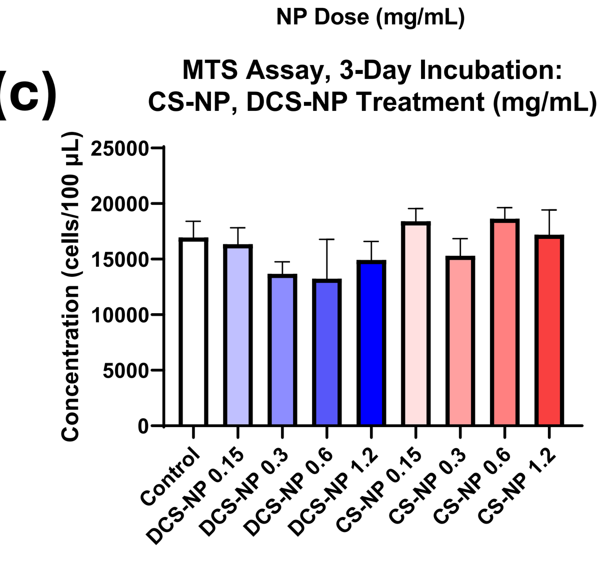
(a)



(b)



(c)



(d)

Treatment	p, ANOVA
BNP	0.6022
CS-NP, DCS-NP, 1-Day	0.2670
CS-NP, DCS-NP, 3-Day	<0.0001

(e)

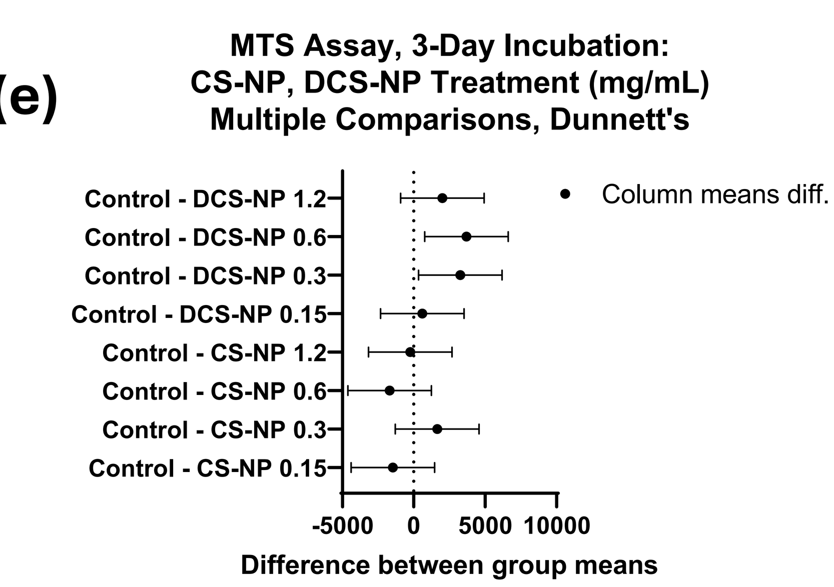


Figure 5: MTS Metabolic Activity Assay. Figure shows mean cell concentration after BNP treatment (a), CS-NP and DCS-NP treatment (b), CS-NP and DCS-NP three-day (repeated exposure) treatment (c), summary of experiment-wise between-group one-way ANOVAs (d), and post-hoc multiple comparisons (e)

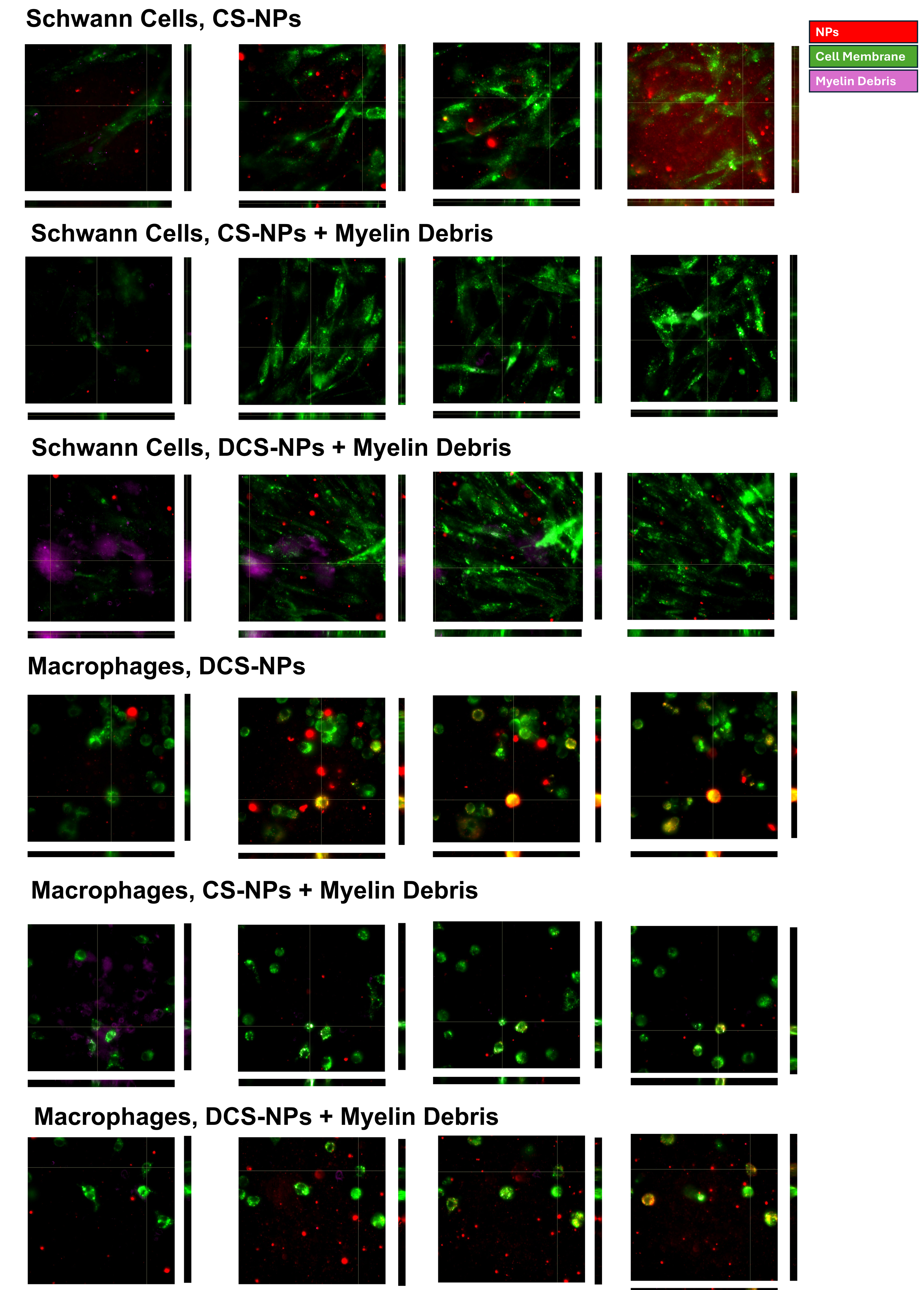


Figure 6: Live Cell Imaging. Z-stack images (confocal, 63x) shown at times 0, 6, and 12 hours for each condition from left to right. All NPs at 0.6 mg/mL. All myelin debris at 0.6 mg/mL. See legend, individual plot descriptions above

Conclusion and Future Directions

- SCs do not appear to internalize BNPs with two-hour incubation. Extracellular agglomerates possibly seen at higher concentrations
- Analysis suggests one-day incubation of BNPs, CS-NPs, or DCS-NPs is not a significant effector of cell viability. One-way ANOVA of extended incubation suggests that extended incubation with DCS-NPs has a significantly negative effect on cell viability (p<0.0001)
- Live cell imaging suggests myelin presence at 0.6 mg/mL does not affect LPS-induced macrophage NP uptake. SC uptake of NPs is not evident both with and without MD. Future direction could involve quantifying percent uptake.

References and Acknowledgements

References:
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